Continuous Time Fourier Series and Properties

Semester 2, 2025/2026



Recap from Week 2

Because of the superposition property, the **response of an LTI system** to any input consisting of a **linear combination of basic signals** is the same **linear combination of the individual responses** to each of the basic signals.

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \qquad x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

The response of an LTI system to a **complex exponential** also has a particularly simple form.

The Eigenfunction Property

Complex exponentials are indeed **eigenfunctions of LTI systems**: the response of an LTI system to a complex exponential input is the same complex exponential with only a change in amplitude:

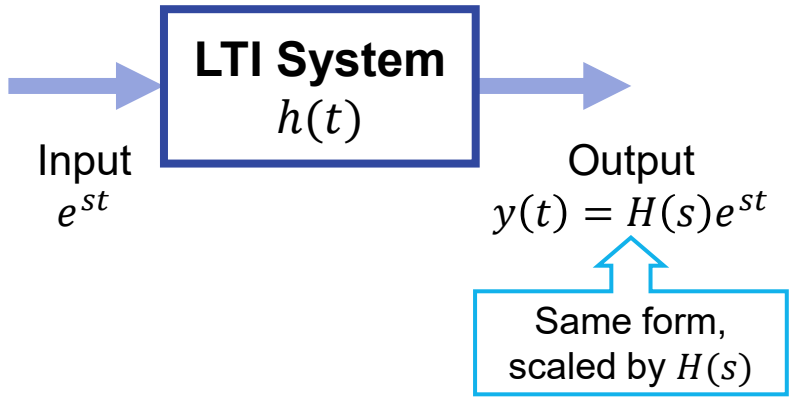
Continuous time: $e^{st} \rightarrow H(s)e^{st}$

Discrete time: $z^n \rightarrow H(z)z^n$

where the complex amplitude factor $H(s)$ or $H(z)$ will in general be a function of the complex variable s or z.

Eigenfunction - A signal for which the system output is a (possibly complex) constant times the input

Eigenvalue – The amplitude factor



The constant $H(s)$ for a specific value of s is the eigenvalue associated with the eigenfunction e^{st}.

The factor $H(s)$ or $H(z)$ is called the **system's frequency response**, and it converts **convolution** (time domain) into **multiplication** (s-domain).

Decomposing Signals in terms of Eigenfunctions

- If the input to an LTI system is represented as a linear combination of complex exponentials, then the output can also be represented as a **linear combination of the same complex exponential signals**.
- Each coefficient in this representation of the output is obtained as the product of the corresponding coefficient a_k of the input and the system's eigenvalue $H(s_k)$ associated with the eigenfunction $e^{s_k t}$.

Let $x(t)$ correspond to a linear combination of three complex exponentials:

$$x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + a_3 e^{s_3 t}$$

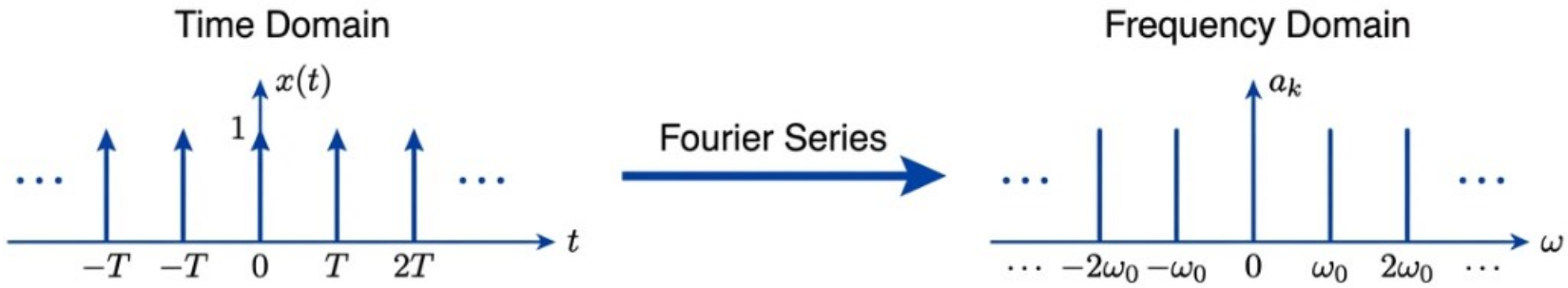
From the eigenfunction property, the response to each separately is

$$a_1 e^{s_1 t} \rightarrow a_1 H(s_1) e^{s_1 t}$$
$$a_2 e^{s_2 t} \rightarrow a_2 H(s_2) e^{s_2 t}$$
$$a_3 e^{s_3 t} \rightarrow a_3 H(s_3) e^{s_3 t}$$

and from the superposition property the response to the sum is the sum of the responses, so that

$$y(t) = a_1 H(s_1) e^{s_1 t} + a_2 H(s_2) e^{s_2 t} + a_3 H(s_3) e^{s_3 t}$$

Fourier Series Representation



Concept: Represent periodic signal as weighted sum of complex exponentials.

In week 1, we introduced two basic periodic signals:

- Sinusoidal signal $x(t) = \cos \omega_0 t$
- Periodic complex exponential $x(t) = e^{j\omega_0 t}$

Both of these signals are periodic with fundamental frequency ω_0 and fundamental period $T = \frac{2\pi}{\omega_0}$.

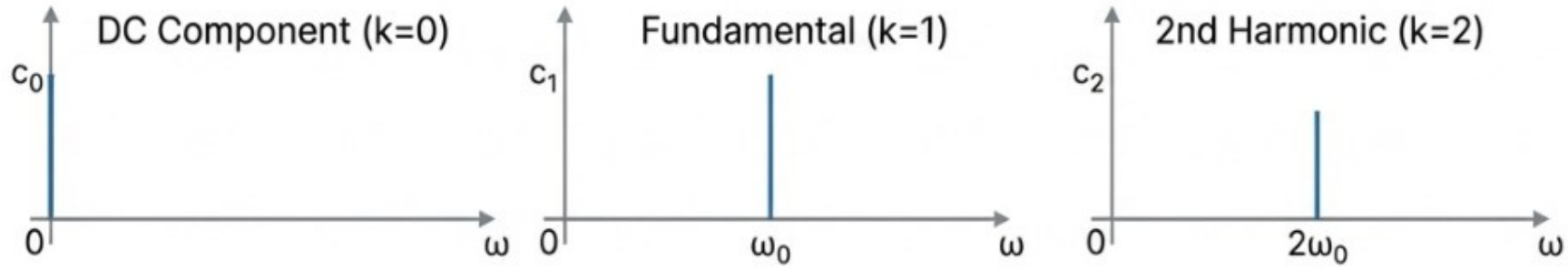
Linear Combination of Harmonically Related Complex Exponentials

Suppose that we have a set of harmonically-related complex exponentials of the form:

$$\phi_k(t) = e^{jk\omega_0 t} = e^{jk(2\pi/T)t}, \qquad k = 0, \pm 1, \pm 2, \ldots$$

The fundamental frequency of the kth complex sinusoid ϕ_k is $k\omega_0$, an integer multiple of ω_0. Each of these signals is periodic with period T.

Harmonics



Linear Combination of Harmonically Related Complex Exponentials

We can represent a periodic complex-valued function $x(t)$ as a linear combination of these harmonics as:

Synthesis Equation

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \underbrace{\sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}}_{\text{Fourier Series representation}}$$

The variables a_k represent the complex weight of each harmonic component.

Example 3.1

Consider a periodic signal $x(t)$, with fundamental frequency 2π, that is expressed as

$$x(t) = \sum_{k=-3}^{3} a_k e^{jk2\pi t}$$

where $a_0 = 1$, $a_1 = a_{-1} = \frac{1}{4}$, $a_2 = a_{-2} = \frac{1}{2}$, $a_3 = a_{-3} = \frac{1}{3}$.

$$x(t) = \frac{1}{3}e^{-j6\pi t} + \frac{1}{2}e^{-j4\pi t} + \frac{1}{4}e^{-j2\pi t} + 1 + \frac{1}{4}e^{j2\pi t} + \frac{1}{2}e^{j4\pi t} + \frac{1}{3}e^{j6\pi t}$$

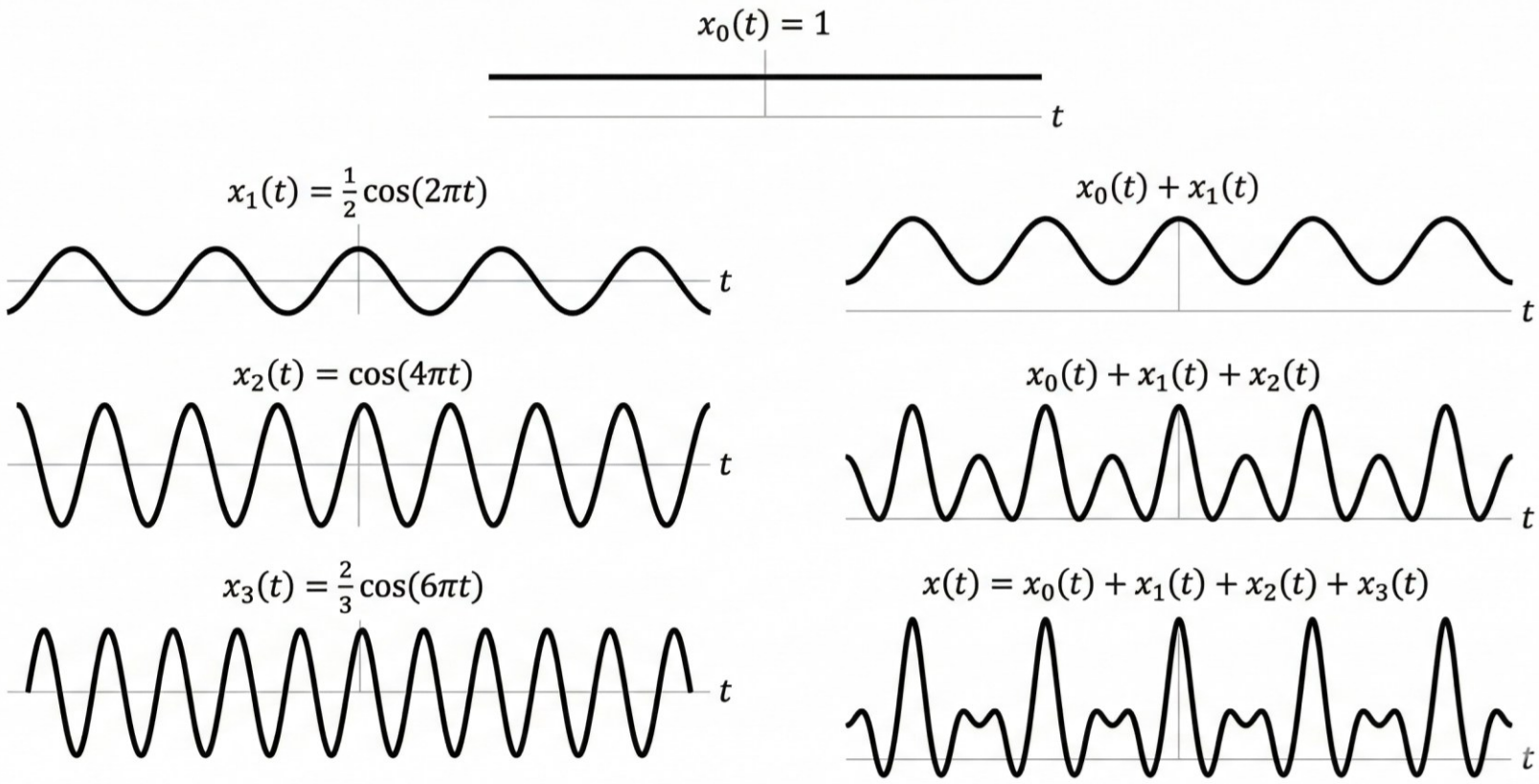
$$x(t) = 1 + \frac{1}{4}\left(e^{j2\pi t} + e^{-j2\pi t}\right) + \frac{1}{2}\left(e^{j4\pi t} + e^{-j4\pi t}\right) + \frac{1}{3}\left(e^{j6\pi t} + e^{-j6\pi t}\right)$$

Using Euler's relation $\left(\cos x = \frac{e^{ix}+e^{-ix}}{2}\right)$, we can write

$$x(t) = 1 + \frac{1}{2}\cos 2\pi t + \cos 4\pi t + \frac{2}{3}\cos 6\pi t$$

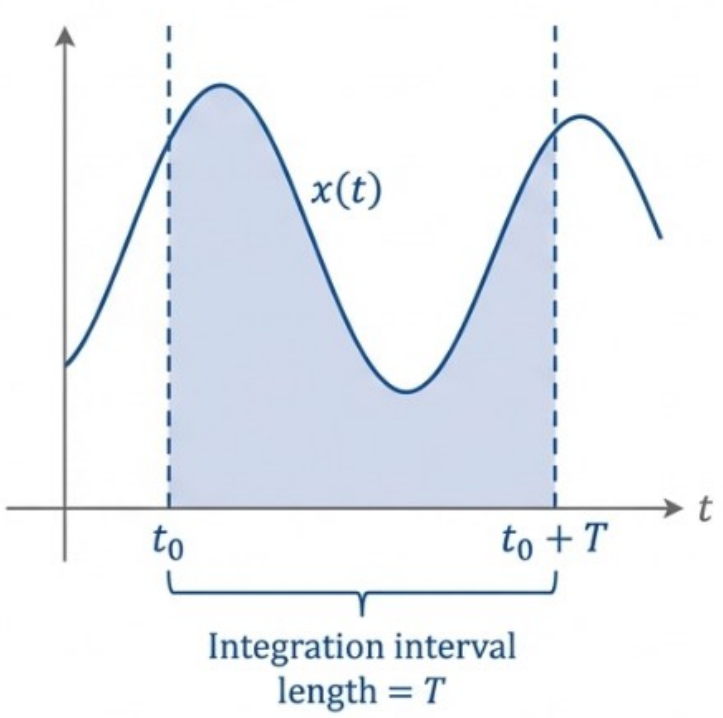
Example 3.1 (cont'd)

Construction of the signal $x(t)$ as a linear combination of harmonically related sinusoidal signals.



Fourier Series Coefficients a_k

- The Fourier series coefficients (or spectral coefficients) a_k measure the portion of the signal at each harmonic.
- Integration is performed over any single interval of length T.
- The FS coefficients are known as *frequency-domain representation* of $x(t)$ because each FS coefficient is associated with a complex sinusoid of a different frequency.



Procedure for Determining the Coefficients a_k

1. Multiply signal by $e^{-jn\omega_0 t}$

$$x(t)e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}\, e^{-jn\omega_0 t}$$

2. Integrate over period T

$$\int_0^T x(t)e^{-jn\omega_0 t}\, dt = \sum_{k=-\infty}^{\infty} a_k \left[\int_0^T e^{j(k-n)\omega_0 t}\, dt\right]$$

$$\int_0^T x(t)e^{-jn\omega_0 t}\, dt = Ta_n$$

$$\int_0^T e^{j(k-n)\omega_0 t}\, dt = \begin{cases} T, & k = n \\ 0, & k \neq n \end{cases}$$

Analysis Equation

$$a_n = \frac{1}{T}\int_T x(t)e^{-jn\omega_0 t}\, dt$$

Fourier Series of Periodic CT Signal

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

where

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_T x(t)e^{-jk(2\pi/T)t}\,dt$$

Setting $k = 0$ yields

$$a_0 = \frac{1}{T}\int_T x(t)\,dt$$

a_0 is the dc component of $x(t)$ or average value of $x(t)$ over one period.

Trigonometric Fourier Series

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

The coefficient a_k can be expressed as $a_k = |a_k| e^{j\theta_k}$.

$$x(t) = a_0 + \sum_{k=1}^{\infty} 2|a_k| \cos(k\omega_0 t + \theta_k)$$

or

$$x(t) = a_0 + \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t) + B_k \sin(k\omega_0 t)$$

$$A_k = \frac{2}{T} \int_0^T x(t) \cos(k\omega_0 t)\, dt$$

$$B_k = \frac{2}{T} \int_0^T x(t) \sin(k\omega_0 t)\, dt$$

Example 3.2: FS Representation by Inspection

Consider the signal

$$x(t) = \sin \omega_o t$$

whose fundamental frequency is ω_0.

For this simple case, it is easier to expand the sinusoidal signal as a linear combination of complex exponentials and identify the Fourier series coefficients by inspection.

Using Euler's formula,

$$\sin \omega_o t = \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t}$$

Example 3.2 (Cont'd)

Equating each term:

$$x(t) = \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

We obtain

$$a_1 = \frac{1}{2j}, \qquad a_{-1} = -\frac{1}{2j}$$

$$a_k = 0, \qquad k \neq +1 \text{ or } -1$$

Example 3.3

Find the Fourier series coefficients for

$$x(t) = 1 + \sin\omega_0 t + 2\cos\omega_0 t + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

Plot the magnitude and phase of the coefficients.

Answer:

$$a_0 = 1$$

$$a_1 = \left(1 + \frac{1}{2j}\right) = 1 - \frac{1}{2}j$$

$$a_{-1} = \left(1 - \frac{1}{2j}\right) = 1 + \frac{1}{2}j$$

$$a_2 = \frac{1}{2}e^{j(\pi/4)} = \frac{\sqrt{2}}{4}(1 + j)$$

$$a_{-2} = \frac{1}{2}e^{-j(\pi/4)} = \frac{\sqrt{2}}{4}(1 - j)$$

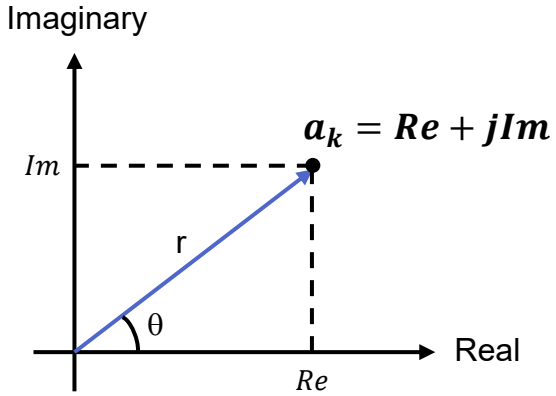
If $z = re^{j\theta}$,
then $z = r\cos\theta + j(r\sin\theta)$

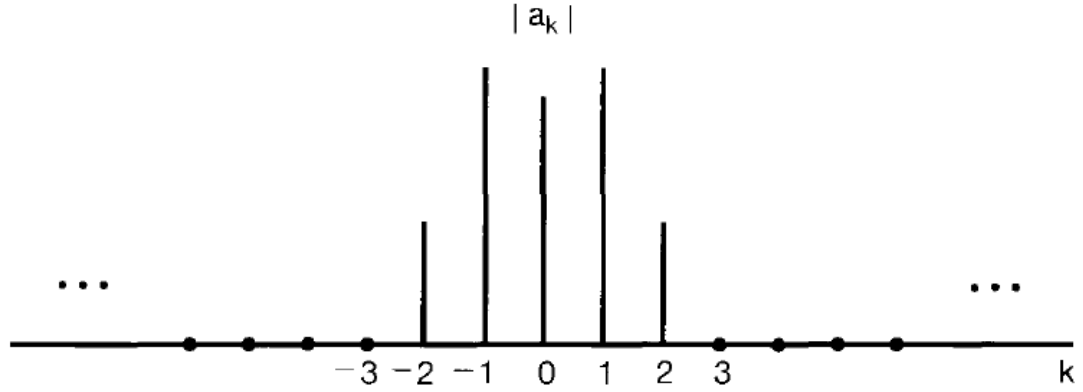
Example 3.3 (Cont'd)

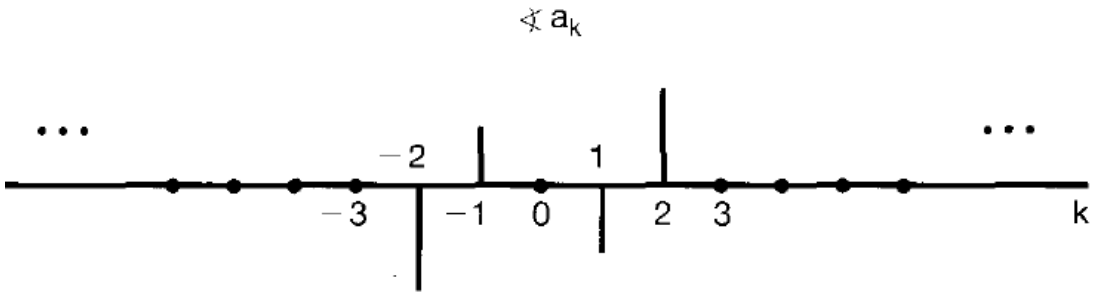
$$a_k = Re + jIm$$

Magnitude: $|a_k| = \sqrt{Re^2 + Im^2}$

Phase: $\measuredangle a_k = \tan^{-1}\left(\frac{Im}{Re}\right)$





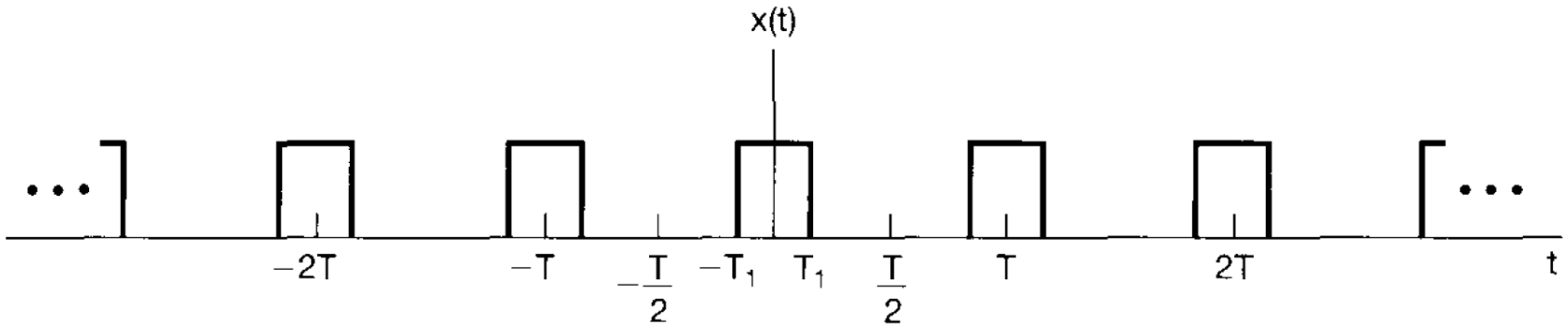


Example 3.4

A periodic square wave is defined over one period as

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

Determine the Fourier series coefficients for $x(t)$.



Because of the symmetry of $x(t)$ about $t = 0$, it is convenient to choose $-T/2 \leq t < T/2$ as the interval over which the integration is performed.

Example 3.4 (Cont'd)

$$a_0 = \frac{1}{T}\int_{-T_1}^{T_1} dt = \frac{2T_1}{T}$$

For k ≠ 0,

$$a_k = \frac{1}{T}\int_{-T_1}^{T_1} e^{-jk\omega_0 t}dt = -\frac{1}{jk\omega_0 T}e^{-jk\omega_0 t}\Big|_{-T_1}^{T_1}$$

$$a_k = \frac{2}{k\omega_0 T}\left[\underbrace{\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{2j}}_{\sin(k\omega_0 T_1)}\right]$$

$$a_k = \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(k\omega_0 T_1)}{k\pi}$$

$\omega_0 T = 2\pi$